

CS57300
PURDUE UNIVERSITY

DATA MINING

NAIVE BAYES CLASSIFIER

- ▶ Simplifying (naive) assumption: attributes are conditionally independent given the class
- ▶ Strengths:
 - ▶ Easy to implement
 - ▶ Often performs well even when assumption is violated
 - ▶ Can be learned incrementally
- ▶ Weaknesses:
 - ▶ Class conditional assumption produces skewed probability estimates
 - ▶ Dependencies among variables cannot be modeled

NBC LEARNING

- ▶ Model space
 - ▶ Parametric model with specific form (i.e., based on Bayes rule and assumption of conditional independence)
 - ▶ Models vary based on parameter estimates in CPDs
- ▶ Search algorithm
 - ▶ MLE optimization of parameters (convex optimization results in exact solution)
- ▶ Scoring function
 - ▶ Likelihood of data given NBC model form

NBC: MAP ESTIMATION

- ▶ Consider a simplified scenario: binary classification (i.e., $L=2$) and each attribute is binary (i.e., $K(j)=2$)
- ▶ Priors: $p_1 \sim \text{Beta}(a, b), q_l^{j1} \sim \text{Beta}(\alpha_l^j, \beta_l^j)$
- ▶ MAP estimate:
 - ▶ Maximize $P(D | \theta)P(\theta)$

NBC: MAP ESTIMATION

$$\begin{aligned}
 P(D | \theta)P(\theta) &= \left(\prod_{i=1}^n \prod_{j=1}^m P(x_{ij} | c_i)P(c_i) \right) \times P(p_1) \times \prod_{l=0}^1 \prod_{j=1}^m P(q_l^{j1}) \\
 &= \prod_{l=0}^1 p_l^{N_l} \prod_{l=0}^1 \prod_{j=1}^m \prod_{k=0}^1 (q_l^{jk})^{N_l^{jk}} \times P(p_1) \times \prod_{l=0}^1 \prod_{j=1}^m P(q_l^{j1})
 \end{aligned}$$



$$p_1 \sim \text{Beta}(a + N_1, b + N_0), q_l^{j1} \sim \text{Beta}(\alpha_l^{j1} + N_l^{j1}, \beta_l^{j1} + N_l^{j0})$$

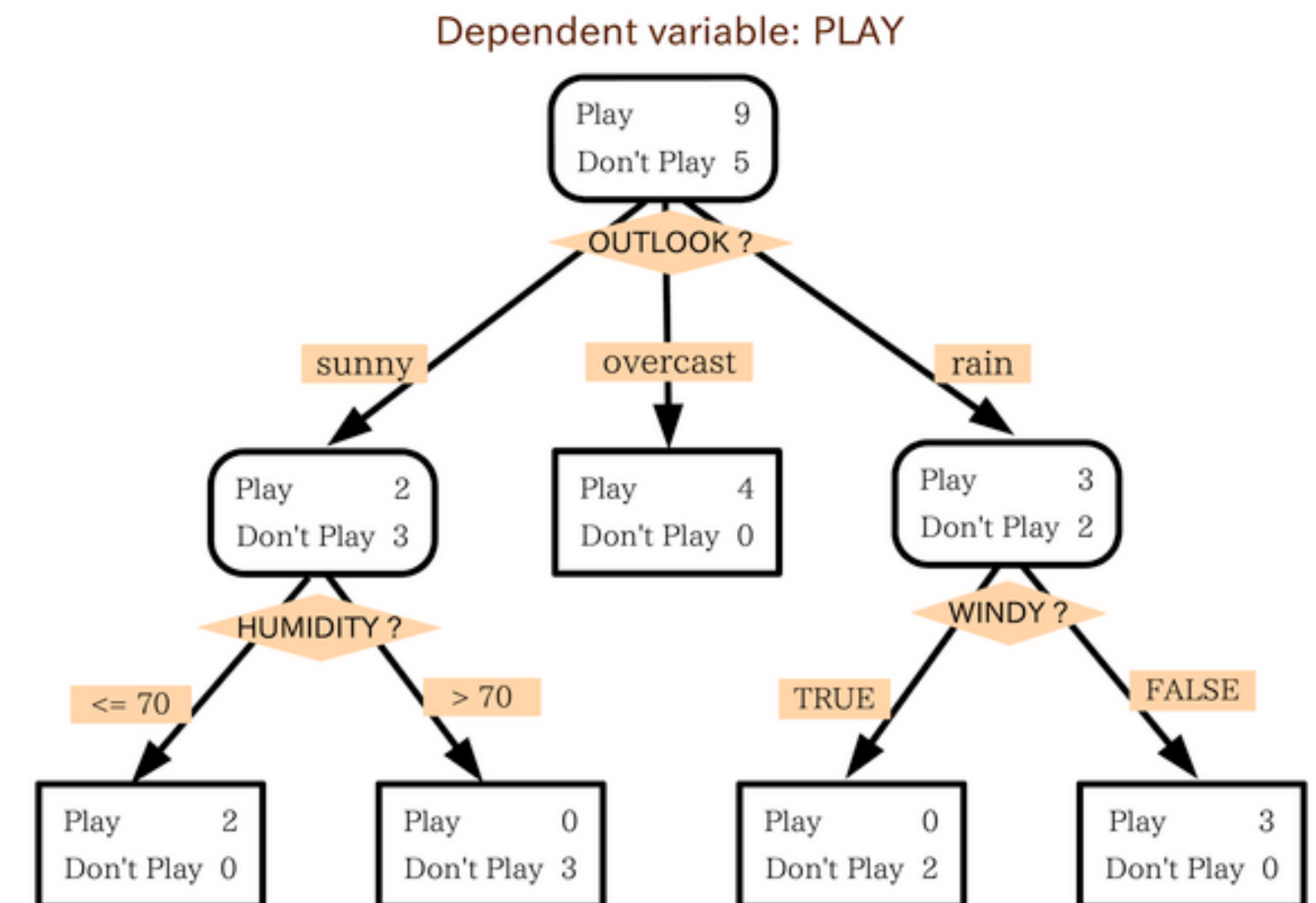


$$[p_1]_{MAP} = \frac{a + N_1 - 1}{a + b + n - 2}, [q_l^{l1}]_{MAP} = \frac{\alpha_l^{j1} + N_l^{j1} - 1}{\alpha_l^{j1} + \beta_l^{j1} + N_l - 2}$$

DECISION TREES

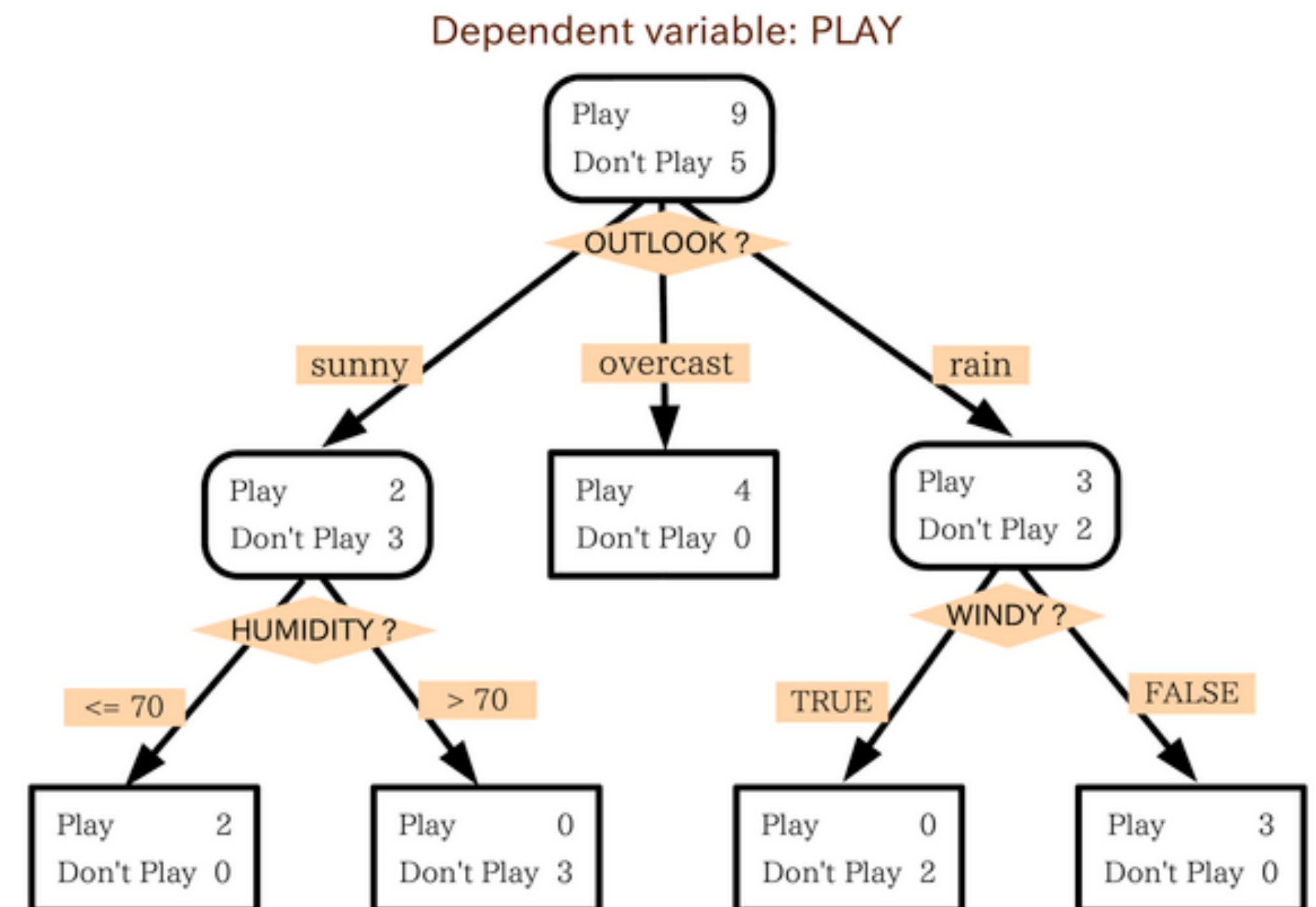
TREE MODELS: KNOWLEDGE REPRESENTATION

- ▶ A decision tree has 2 kinds of nodes
 - ▶ Each internal node is a question on features. It branches out according to the answers
 - ▶ Each leaf node has a class label, determined by the majority vote of training examples reaching that leaf
- ▶ Advantages
 - ▶ Easy inference
 - ▶ Can handle mixed variables
 - ▶ Easy for humans to understand



TREE LEARNING

- ▶ Model space: All possible decision trees
 - ▶ Each layer can include different attributes
 - ▶ Each attribute can split on different values
 - ▶ Can have different number of layers
- ▶ Scoring function: Misclassification rate
- ▶ Search process: Heuristic search
 - ▶ Greedy, recursive divide and conquer

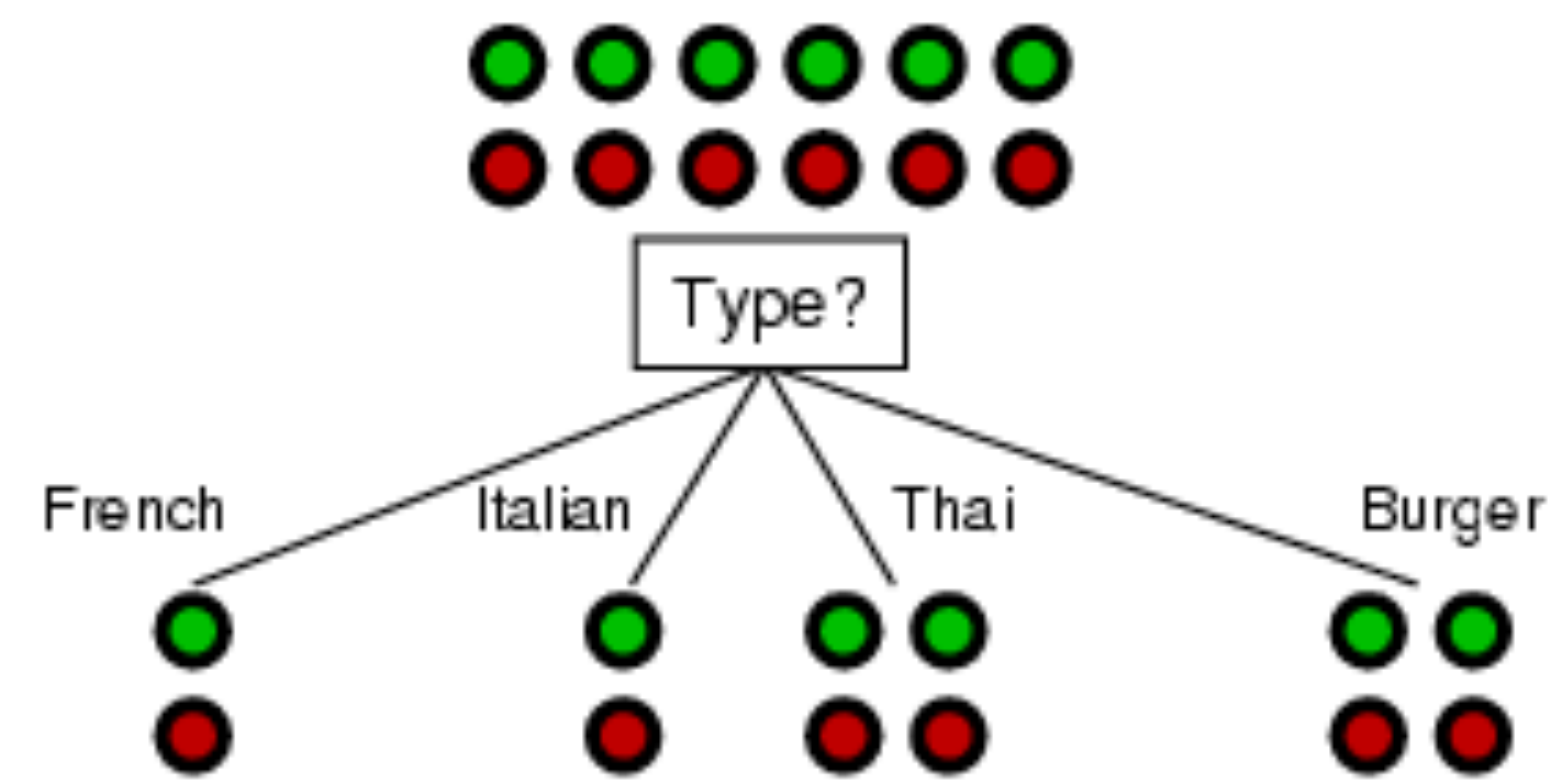
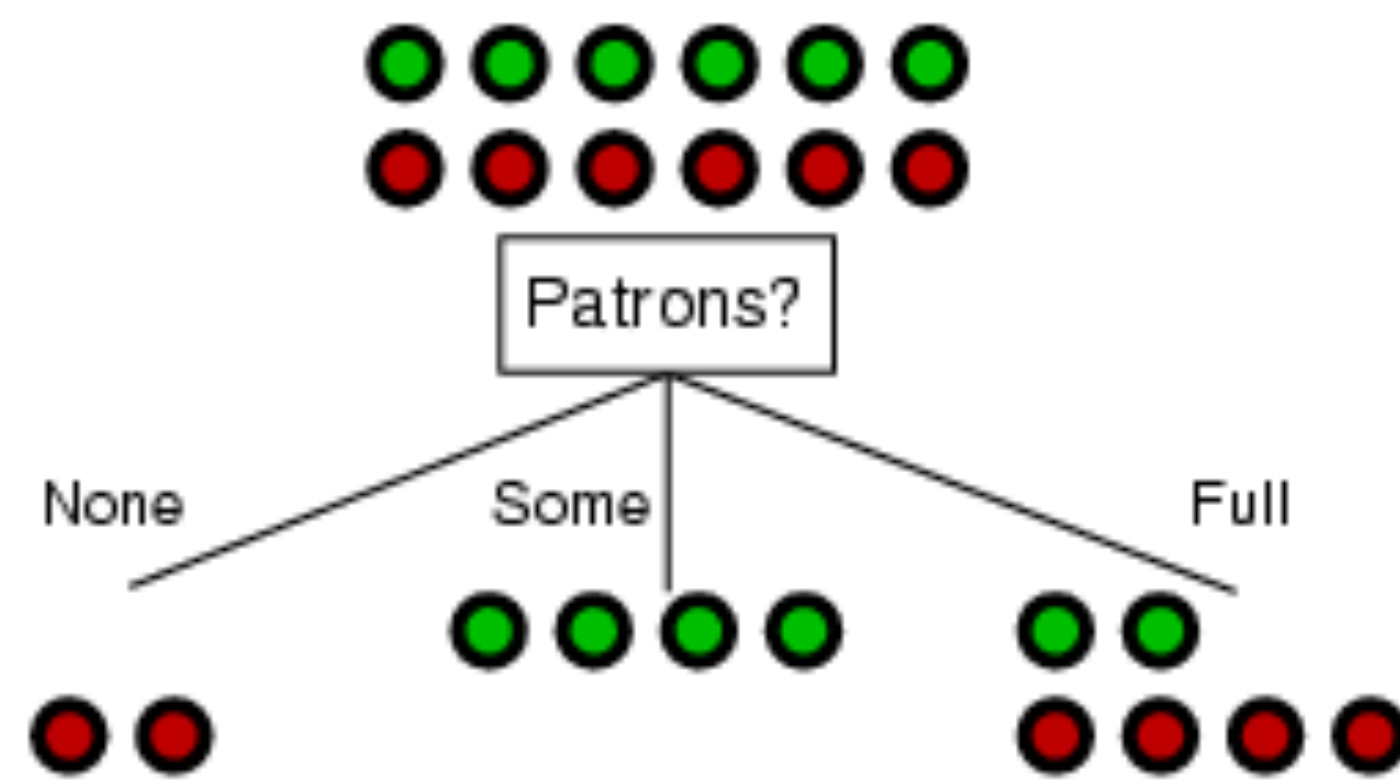


TREE LEARNING

- ▶ Top-down recursive divide and conquer algorithm
 - ▶ Start with all training examples at root
 - ▶ Select **best** attribute/feature: Take a greedy view to decide how “good” an attribute is
 - ▶ Partition examples by selected attribute
 - ▶ Recurse and repeat
- ▶ Other issues:
 - ▶ When to stop growing
 - ▶ Pruning irrelevant parts of the tree

CHOOSING AN ATTRIBUTE/FEATURE

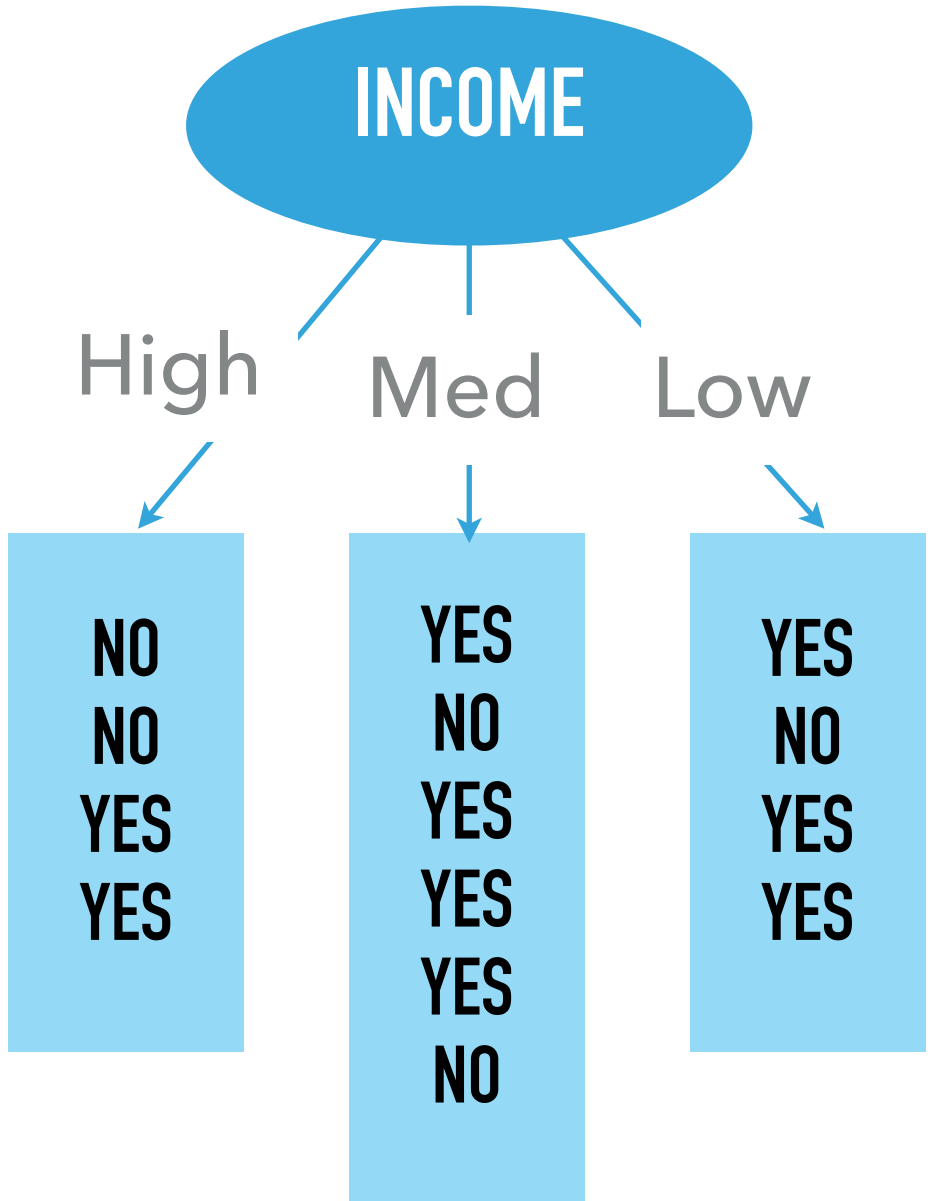
- ▶ Be greedy: choose an attribute that can immediately minimize the misclassification rate (i.e., as if no further subtree will grow)
- ▶ A good feature splits the examples into subsets that distinguish among the class labels as much as possible... ideally into pure sets of "all positive" or "all negative"



ASSOCIATION BETWEEN ATTRIBUTE AND CLASS LABEL

Data

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no



Contingency table

		Class label value	
		Buy	No buy
Attribute value	High	2	2
	Med	4	2
	Low	3	1

A good attribute leads to **highly certain** prediction for training examples sharing the same value on that attribute!

MEASURING UNCERTAINTY: ENTROPY

- ▶ Used to quantify the amount of randomness of a probability distribution.
- ▶ Definition: Suppose a discrete random variable X has a distribution of $P(X)$. The **entropy** $H(X)$ of X is defined by:

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

ENTROPY OF A RANDOM VARIABLE

A completely random binary variable with $P(X)=[0.5,0.5]$ has entropy:

$$H(X) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = -(-0.5 + -0.5) = 1$$

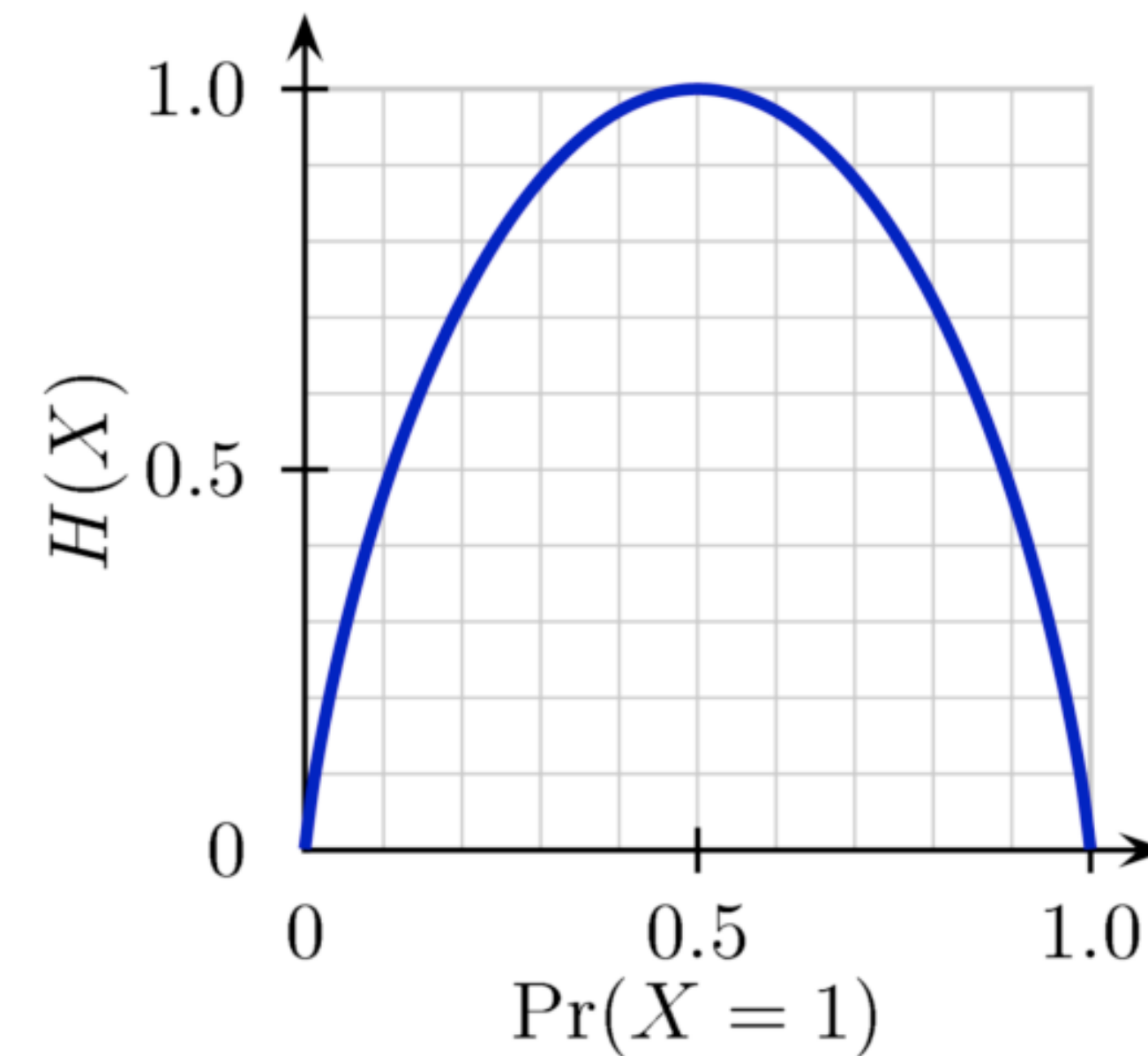
A deterministic variable with $P(X)=[1,0]$ has entropy:

$$H(X) = -(1 \log_2 1 + 0 \log_2 0) = -(0+0) = 0$$

A biased variable with $P(X)=[0.75,0.25]$ has entropy:

$$H(X) = 0.8113$$

The entropy of a probability distribution **p** expresses the **amount of uncertainty** that we have about the values of X



MEASURING CHANGE OF UNCERTAINTY: INFORMATION GAIN

- How much does a feature split decrease the entropy?

$$Gain(S, A) = \underline{Entropy(S)} - \sum_{v \in values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
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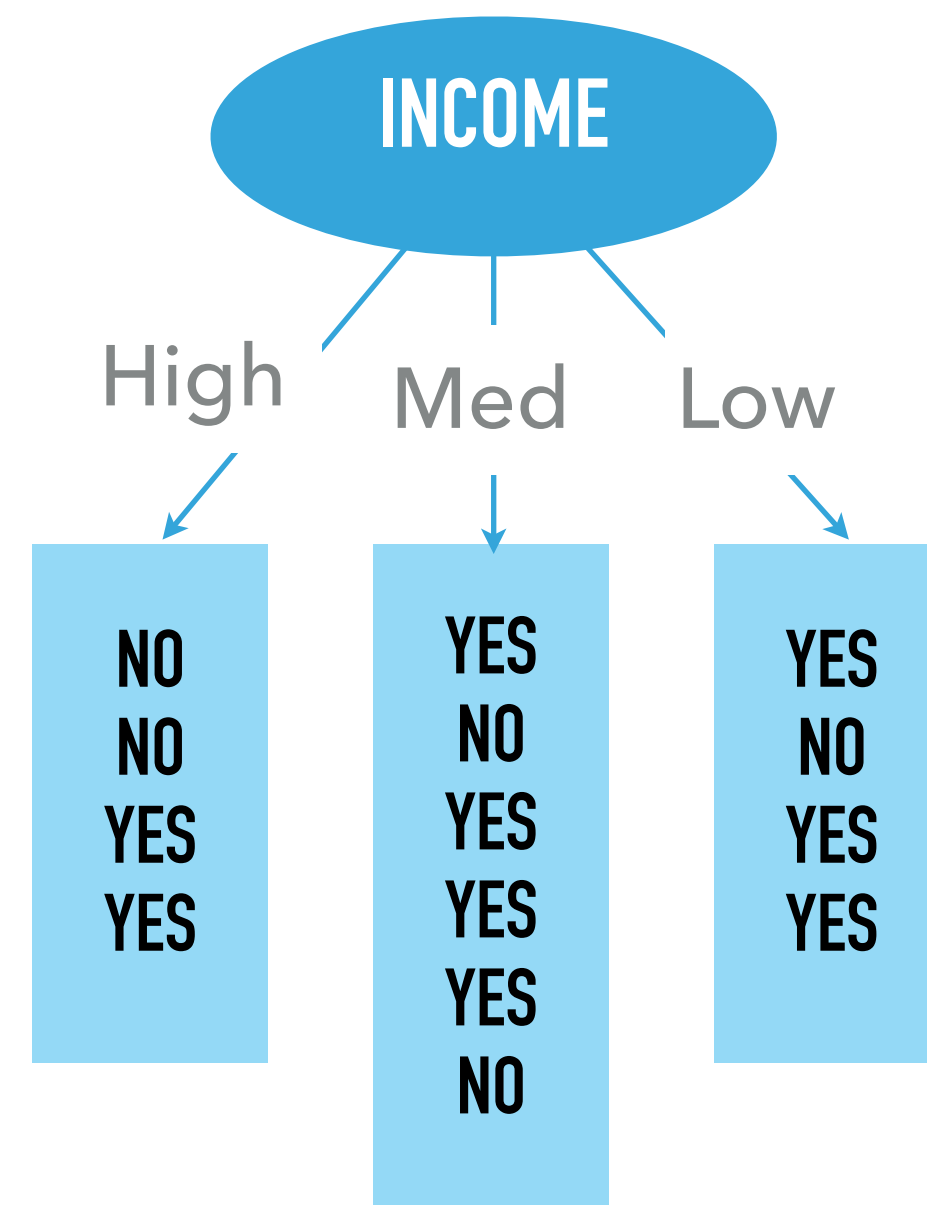
Entropy(S)

$$= -9/14 \log 9/14 - 5/14 \log 5/14$$

$$= 0.9400$$

INFORMATION GAIN

$$Gain(S, A) = Entropy(S) - \sum_{v \in values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$



$$\begin{aligned} Entropy(\text{Income}=\text{High}) \\ = -2/4 \log 2/4 - 2/4 \log 2/4 = 1 \end{aligned}$$

$$\begin{aligned} Entropy(\text{Income}=\text{Med}) \\ = -4/6 \log 4/6 - 2/6 \log 2/6 = 0.9183 \end{aligned}$$

$$\begin{aligned} Entropy(\text{Income}=\text{Low}) \\ = -3/4 \log 3/4 - 1/4 \log 1/4 = 0.8113 \end{aligned}$$

$$\begin{aligned} Gain(S, \text{Income}) \\ = 0.9400 - (4/14 [1] + 6/14 [0.9183] + 4/14 [0.8113]) \\ = 0.029 \end{aligned}$$